Dingqi Zhang

Staff Research Scientist, Multiply Labs

dingqi@berkeley.edu / (+1) 607-379-5156
dz298.github.io/website / Google Scholar

SUMMARY

My research focuses on generalizable robot control: learned feedback policies that reliably execute motion across changes in embodiment, dynamics, and physical interaction. My Ph.D. demonstrated extreme cross-embodiment adaptation in high-frequency quadrotor control. I now extend this to dexterous manipulation, with object-agnostic, contact-driven control, and to humanoid loco-manipulation, with force- and terrain-aware whole-body tracking.

EXPERIENCE

Multiply Labs Jun 2026 – present
Staff Research Scientist San Francisco, CA

- Dexterous manipulation. One object-agnostic, contact-feedback policy tracks trajectories supplied by retargeting or planning across unseen objects on a Wuji hand [2].
- Humanoid whole-body control. A learned tracking controller follows motion references while actively exerting commanded forces on the environment and adapting to varying terrain on a Unitree G1 [1].
- Interaction-informed planning. Object-centric replanning updates humanoid loco-manipulation plans using information revealed through physical interaction [3].
- Mentoring and infrastructure. Advise visiting Ph.D. interns and an external Ph.D. student group; contribute to shared training, sim-to-sim, and hardware-deployment tooling.

RAI Institute Summer 2025
Research Intern, Dexterous Mobile Manipulation Cambridge, MA

- Sim-to-real adaptation for dynamic in-hand skills with model-based RL.

The Chinese University of Hong Kong Summer 2024
Visiting Researcher, host Prof. Ben M. Chen Hong Kong

Zipline Summer 2023
Guidance, Navigation & Control Intern South San Francisco, CA

EDUCATION

University of California, Berkeley 2021 – 2026
Ph.D., Control and Robotics Berkeley, CA

- Thesis: *Learning to Adapt Across Embodiments: Latent-Based Control, Evaluation, and Interpretation for Quadrotors*
- Cross-embodiment control. A single learned 500 Hz feedback controller generalized across quadrotors with orders-of-magnitude differences in dynamics parameters [7, 9].
- Evaluation & interpretation. An open benchmark for adaptive quadrotor control [6] and methods to interpret learned control latents as physical models [5].
- Advisor: Mark W. Mueller · Committee: Jitendra Malik, Koushil Sreenath

Cornell University 2017 – 2021
B.S., Computer Science and Mechanical Engineering Ithaca, NY

- Undergraduate research in autonomous navigation and locomotion with Andy Ruina.

PUBLICATIONS

PREPRINTS & UNDER REVIEW

D. Zhang = me † intern advised by me

[1] BRACE: Adapting Whole-Body References for Force and Terrain Aware Humanoid Motion Tracking
S. Harithas†, C. Yu†, J. Borbon, S. Mondal, W. Zha, S. Sridhar, **D. Zhang**, J. Wang
arXiv preprint, 2026 arXiv project code video

[2] OATH: Object-Agnostic Tracking for Dexterous Hands, Enabling Rapid Physics-based Retargeting
S. Sigurdson†, P. Sriganesh†, J. Borbon, J. Burdick, **D. Zhang**, J. Wang
Under review, 2026

[3] Belief-Overlap Partial Inversion: Object-Centric Flow Replanning for Humanoid Loco-Manipulation
C. Yu†, S. S. Harithas†, J. Borbon, **D. Zhang**, J. Wang
Under review, 2026 project

[4] KinoGen: Customizable Humanoid Loco-Manipulation References via Kinematically Grounded Video Generation
Q. Wu, L. Zhao, S. Hahn, R. Yennu, E. Yang, **D. Zhang**, T. Pang, J. Wang, K. Fang
Under review, 2026 project

[5] Interpreting Control Latents for System Identification via Conditional Flow Matching
D. Zhang, R. Zhang, M. W. Mueller
arXiv preprint, 2026 arXiv

PUBLISHED

[6] A Simulation Evaluation Suite for Robust Adaptive Quadcopter Control
D. Zhang, R. Tao, S. Cheng, N. Hovakimyan, M. W. Mueller
American Control Conference (ACC), 2026 arXiv code

[7] A Learning-Based Quadcopter Controller With Extreme Adaptation
D. Zhang, A. Loquercio, J. Tang, T.-H. Wang, J. Malik, M. W. Mueller
IEEE Transactions on Robotics (T-RO), 2025 arXiv code video

[8] ProxFly: Robust Control for Close Proximity Quadcopter Flight via Residual Reinforcement Learning
R. Zhang, **D. Zhang**, M. W. Mueller
IEEE International Conference on Robotics and Automation (ICRA), 2025 arXiv video

[9] Learning a Single Near-Hover Position Controller for Vastly Different Quadcopters
D. Zhang, A. Loquercio, X. Wu, A. Kumar, J. Malik, M. W. Mueller
IEEE International Conference on Robotics and Automation (ICRA), 2023 arXiv project code

MENTORING

Students I advise at Multiply Labs, from project scoping through experiments and papers.

Sudarshan Harithas, Brown University — force-aware humanoid tracking [1] — 2026

Chen Yu, Northwestern University — adaptive humanoid loco-manipulation [3] — 2026

Solvin Sigurdson, Caltech — dexterous physics-based retargeting [2] — 2026

Prasanna Sriganesh, Carnegie Mellon University — force-guided search under unknown dynamics — 2026

Ph.D. student group, University of New Mexico — humanoid whole-body control on the G1 — 2026

TALKS

Control Seminar, UC Berkeley — A learning-based quadcopter controller for extreme adaptation — Apr 2025

Bay Area Robotics Symposium, UC Berkeley — poster & talk — 2022, 2024

The Chinese University of Hong Kong; Hong Kong Polytechnic University — Jul 2024

SERVICE & TEACHING

Reviewer — T-RO, RA-L, T-MECH, RSS, ICRA, IROS

Teaching — TA, ME136/236 Dynamics & Control of Autonomous Flight, UC Berkeley (2023, 2024); TA, Master of Future Energy Systems & Technology, DEWA & UC Berkeley (2023–24)

AWARDS & PRESS

Ignite Grant, Jacobs Institute Innovation Catalysts, UC Berkeley — top 3% — 2024

Spark Grant, Jacobs Institute Innovation Catalysts, UC Berkeley — top 7% — 2023

Both for *Tennie*, an autonomous tennis-ball collecting robot (with J. Tang).

Graduate Division Block Grant Award, UC Berkeley — 2021

Featured in *IEEE Spectrum* Video Friday — link — 2024

SKILLS

Methods — RL & imitation learning, sim-to-real, teacher–student distillation, residual RL, flow matching, latent adaptation, physics-based retargeting

Robots — Wuji, Allegro & Dex3 hands; Robotiq grippers; UR5e arms; Unitree G1; quadrotors

Software — Python, C/C++, PyTorch, Isaac Lab, mjlab, MuJoCo, ROS